Landmarks And Features - Abbreviated Version

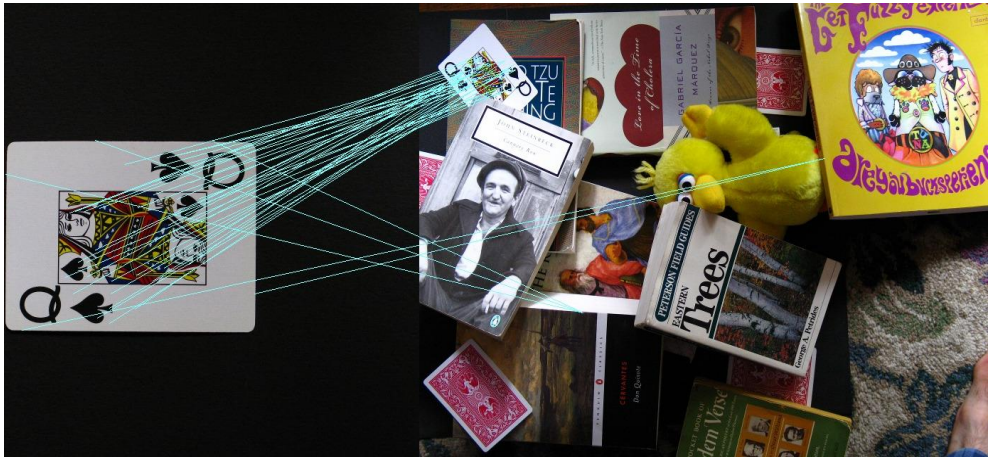
CS 4330

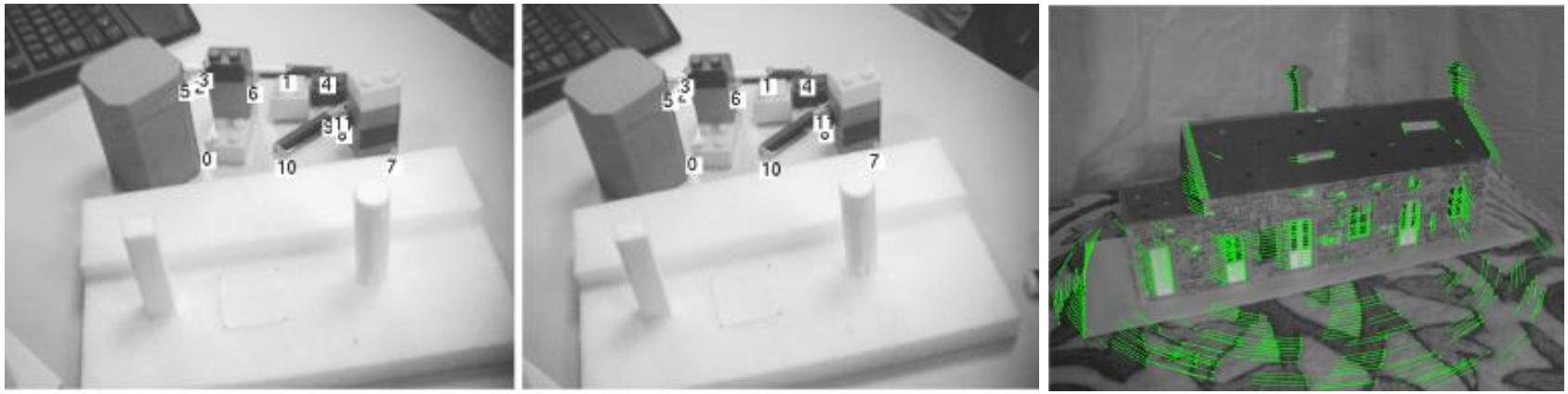
Mathias Kölsch

Some slides from O. Camps, S. Seitz, R. Szeliski, David Lowe, Tom Duerig

Definition and Terminology

- *Feature* is ambiguous but we're stuck with it
- *Landmark* if longer-lived
- *Interest point*





(a) (b)

Features… Why? How?

- What are feature points good for?
  - Motion tracking (robot navigation)
  - Image alignment/registration (stitching, mosaics)
  - Stereo (depth), esp. wide baseline matching
  - 3D reconstruction, structure from motion
  - Object recognition
  - Indexing and database retrieval
- What exactly are features?
  - Gray-scale spatial appearance (2D)
  - Color variation in 2D
  - Motion signatures (spatial and temporal, 3D)
  - 3D structure (point clouds)

Advantages of Features

Use *local* appearance:

PARTS and *RELATIONS*

- Can model general-form objects
- Can deal with occlusion and clutter
- Can handle translation and scaling
- It is robust to segmentation problems

Topic Overview

Introduction

Feature…

- Desired Characteristics
- Detection (extraction)
- Description
- Matching
- Tracking

Reading: Szeliski Chapter 7.1

We Want Invariance!

- Good features should be robust to all sorts of visual changes that can occur between images of the same scene
- Suppose you rotate the image by some angle: Will you still pick up the same features?
- What if you change the brightness?
- Scale?

Types of invariance

- Illumination

Types of invariance

- Illumination
- Scale

Types of invariance

- Illumination
- Scale
- Rotation

Types of invariance

- Illumination
- Scale
- Rotation
- Affine

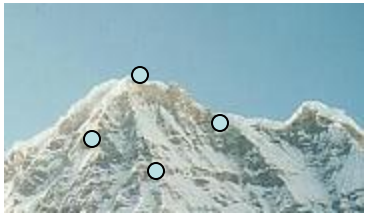
this is not an affine transformation →

Types of invariance

- Illumination
- Scale
- Rotation
- Affine
- Full Perspective

Repeatability

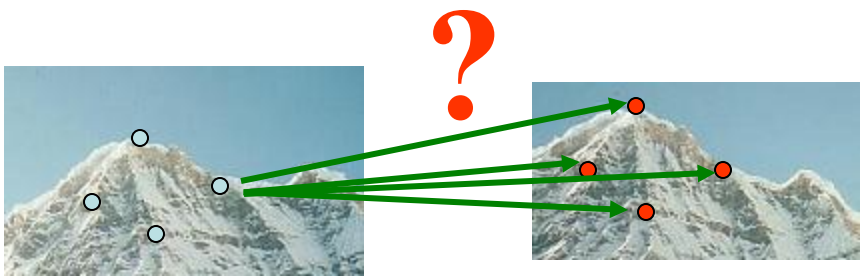
- Detect the *same* points in both images



Feature detection must be repeatable

“Good” Description

- Match features correctly between images: “good” for matching in both directions



We need a good description of the features

What makes a good feature?

See Courses/CS4330/FlowerGardenSequence/*.jpg

Snoop demo

We Want Uniqueness!

Look for image regions that are unusual

- Lead to unambiguous matches in other images

How to define “unusual”?

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Introduction

Feature…

- Desired Characteristics
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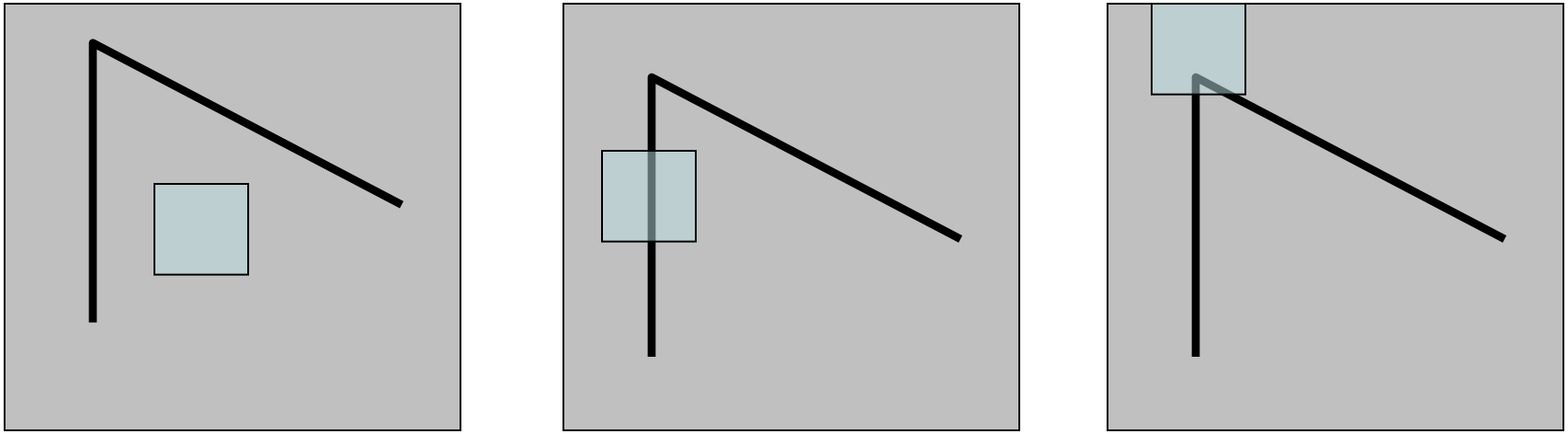
Finding Unusual Spots

Local measures of uniqueness

Suppose we only consider a small window of pixels

- What defines whether a feature is a good or bad candidate?

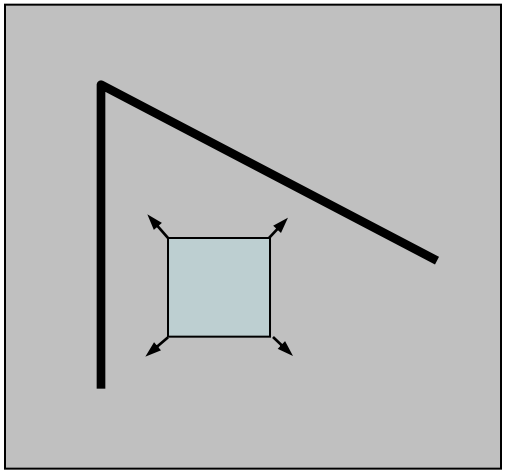


Slide adapted from Darya Frolova, Denis Simakov, Weizmann Institute.

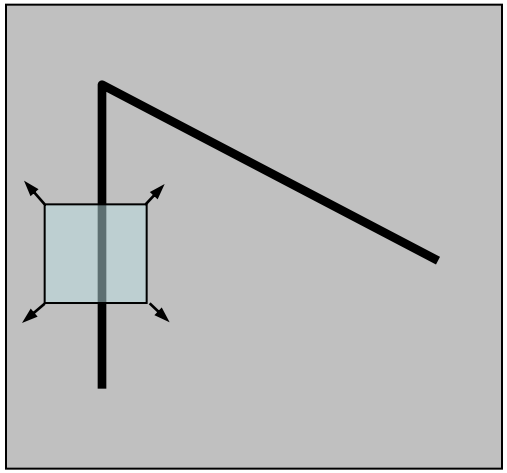
Feature detection for uniqueness

Local measure of feature uniqueness

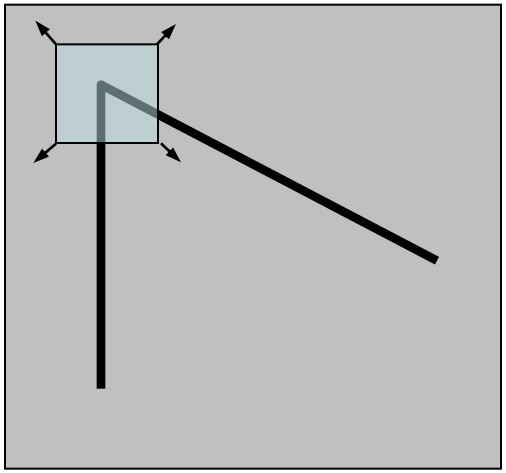
- How does the window change when you shift it?
- Shifting the window in *any direction* causes a *big change*



"flat" region:
no change in all directions

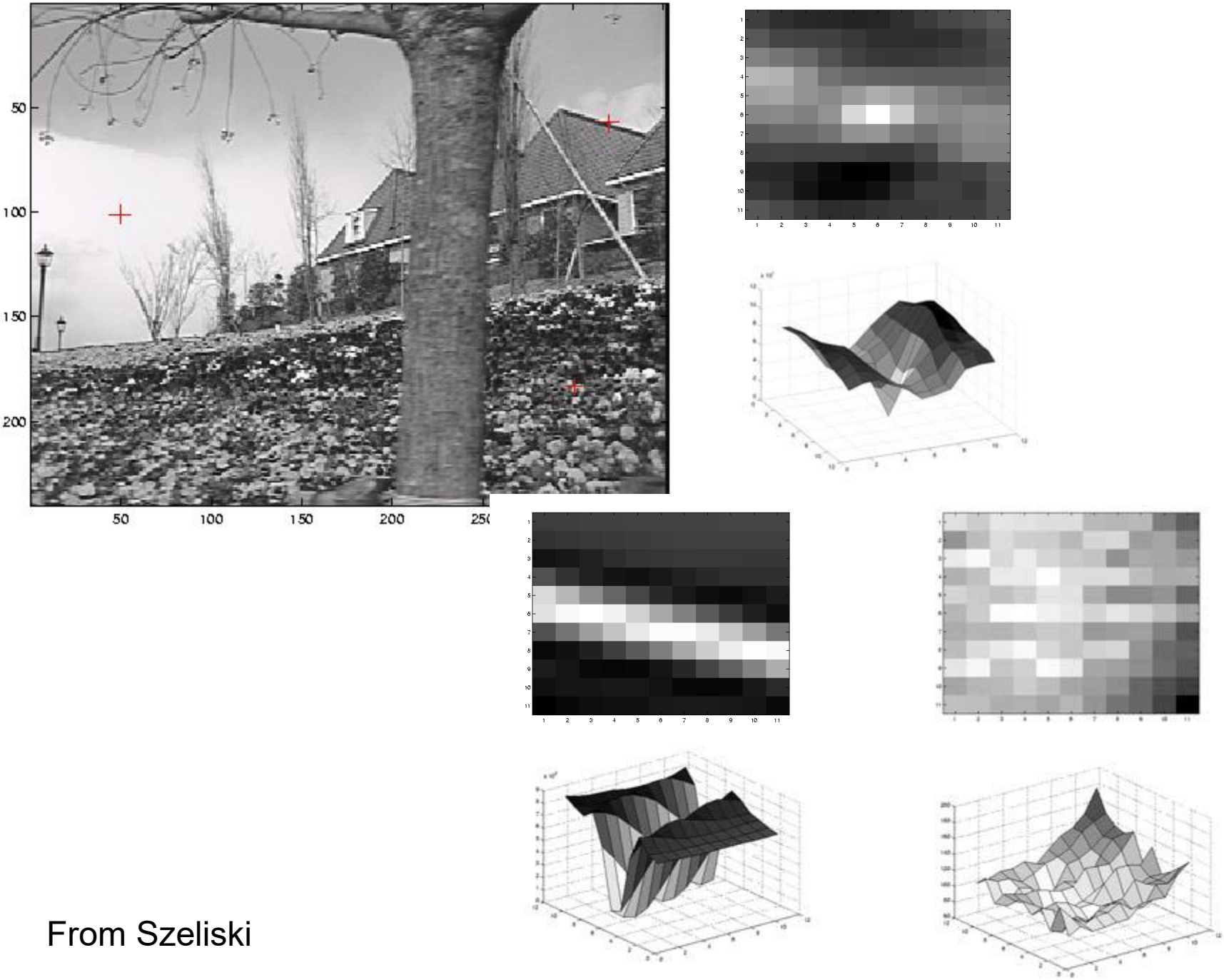


"edge":
no change along the edge direction



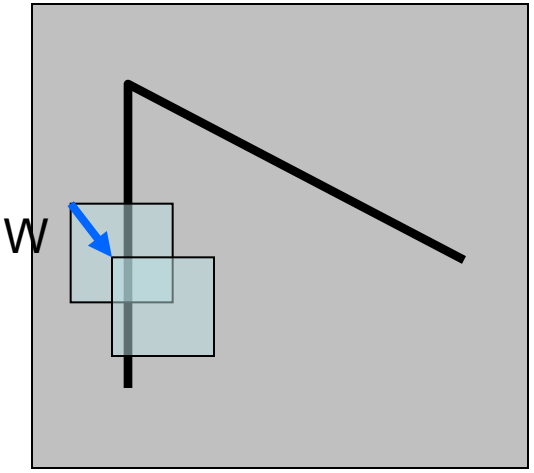
"corner":
significant change in all directions

Slide adapted from Darya Frolova, Denis Simakov, Weizmann Institute.



From Szeliski

Computation of Suitability



Consider shifting the window W by a "displacement" (u,v)

- how do the pixels in W change?
- compare each pixel before and after by summing up the squared differences (SSD)
- this defines an SSD "error" of *E(u,v)*:

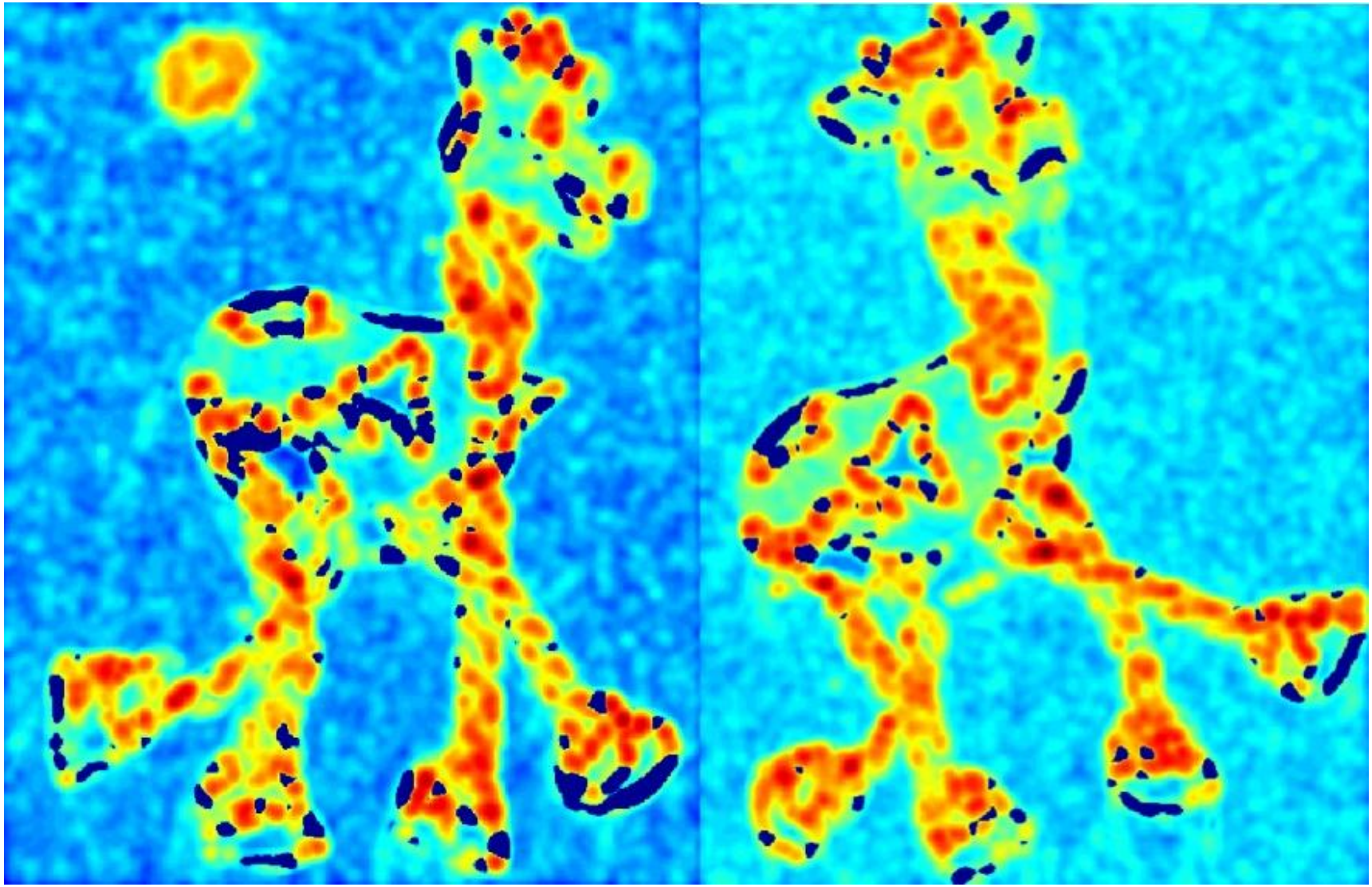
$$E(u,v) = \sum_{(x,y)\in W} [I(x+u, y+v) - I(x,y)]^2$$

Harris & Stephens, 1988: A Combined Corner and Edge Detector

- Also called the auto-correlation function (comparing W with its neighborhood)
- Multiple ways to calculate an approximation of this error and hence "uniqueness" of W: eigenvectors, Hessian matrix, Harris operator…

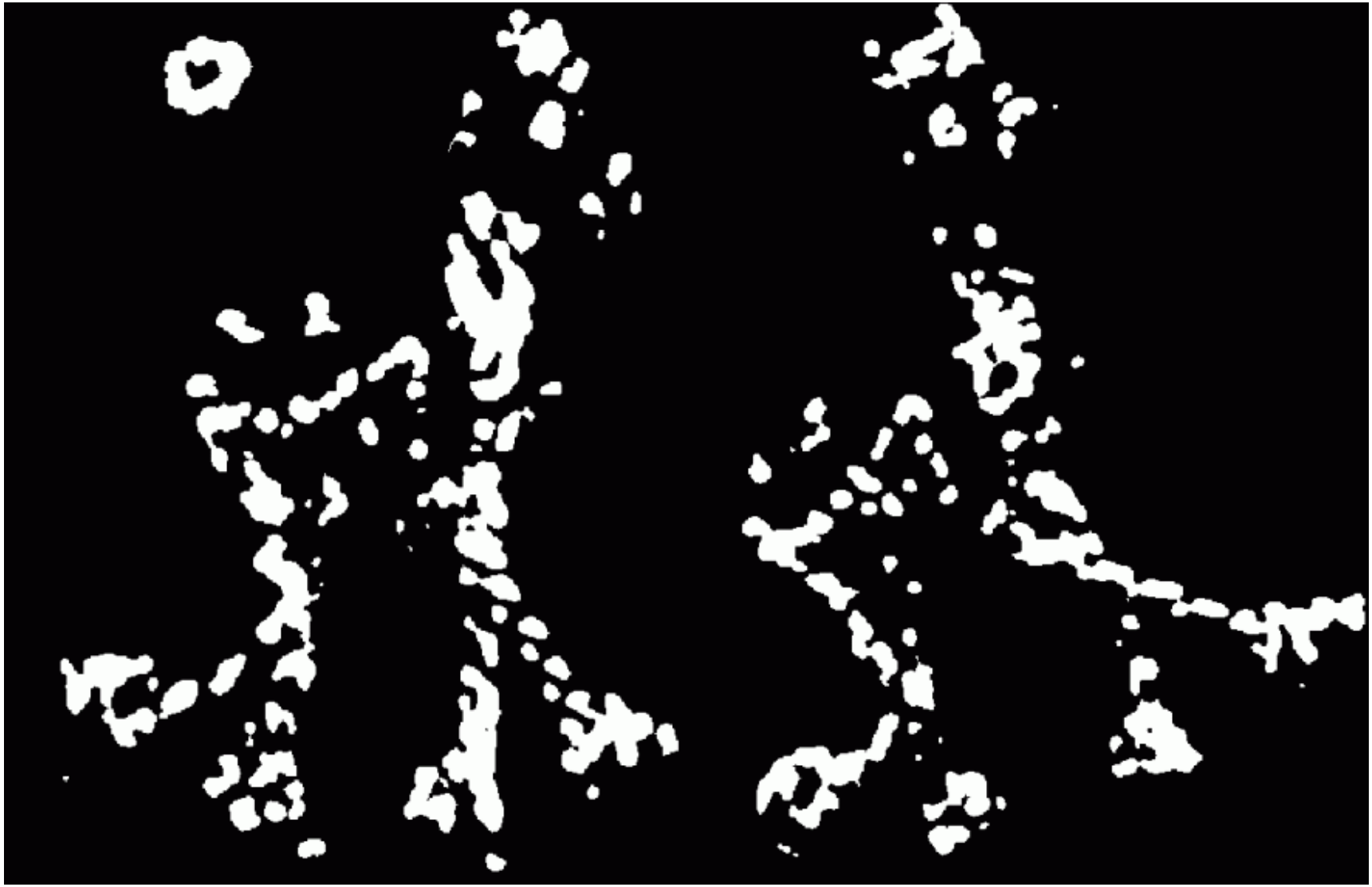
Harris detector example

f value (red high, blue low)

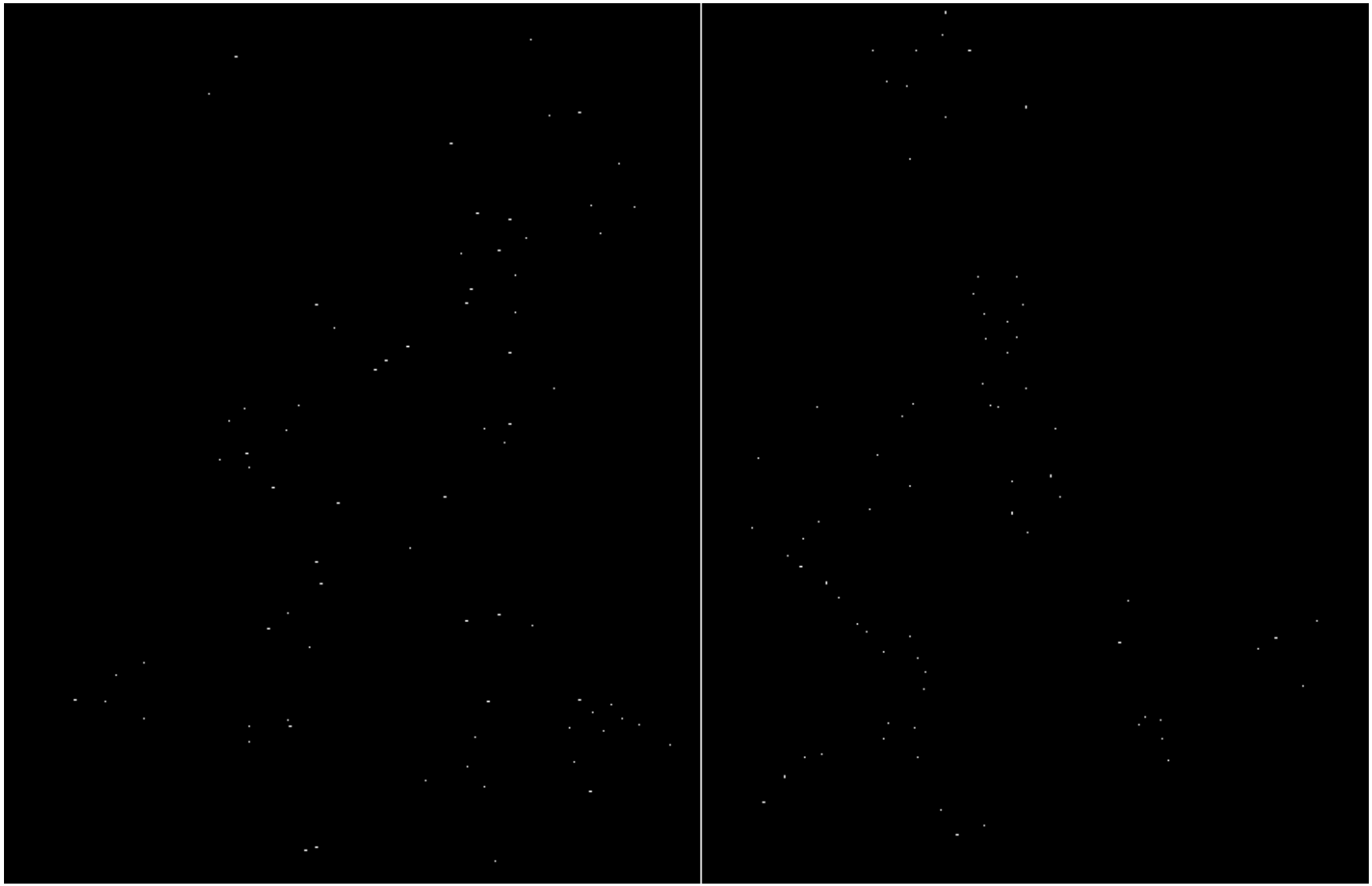


See: https://docs.opencv.org/4.x/dc/d0d/tutorial_py_features_harris.html

Threshold (f > value)



Find local maxima of f



Harris features (in red)

Scale-Space Detection

Finding features in “spatial space” (x-y) does not determine their size. Size needs to consider the so-called “scale space.”

- *Scale-space* features either exist across several scales, or an inherent scale can be determined.
- Use Gaussian & Laplacian pyramids (also called Difference of Gaussians, DoG)
- If the feature is repeatedly present in DoG levels it is scale invariant

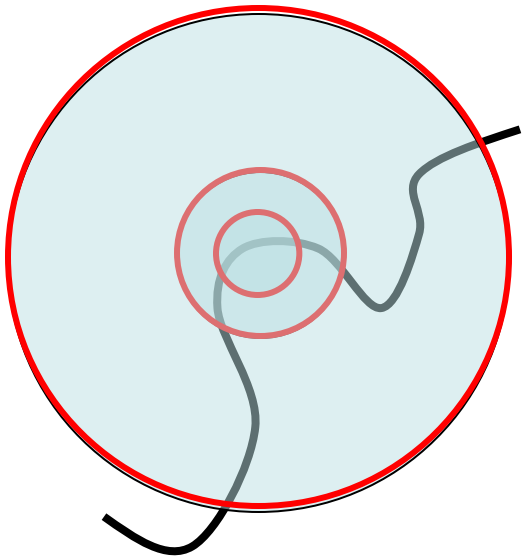
Laplacian Pyramid (Differences Of Gaussians)

Gaussian Pyramid

Laplacian Pyramid

Scale invariant detection

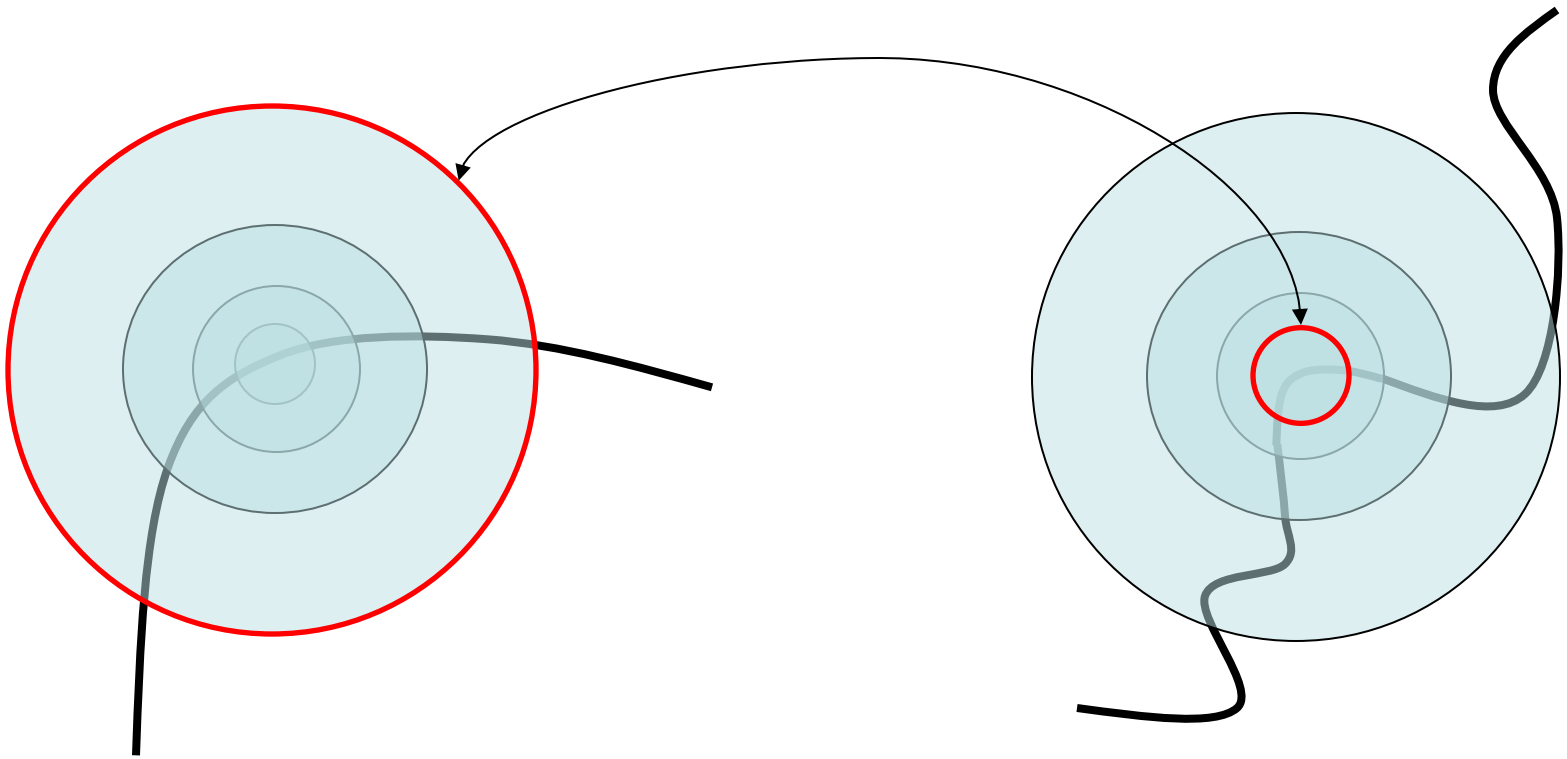
Suppose you're looking for corners



Key idea: find scale that gives local maximum of f

- f is a local maximum in both position and scale
- Common definition of f: Laplacian
  (or difference between two Gaussian filtered images with different sigmas)

Scale Invariant Detection

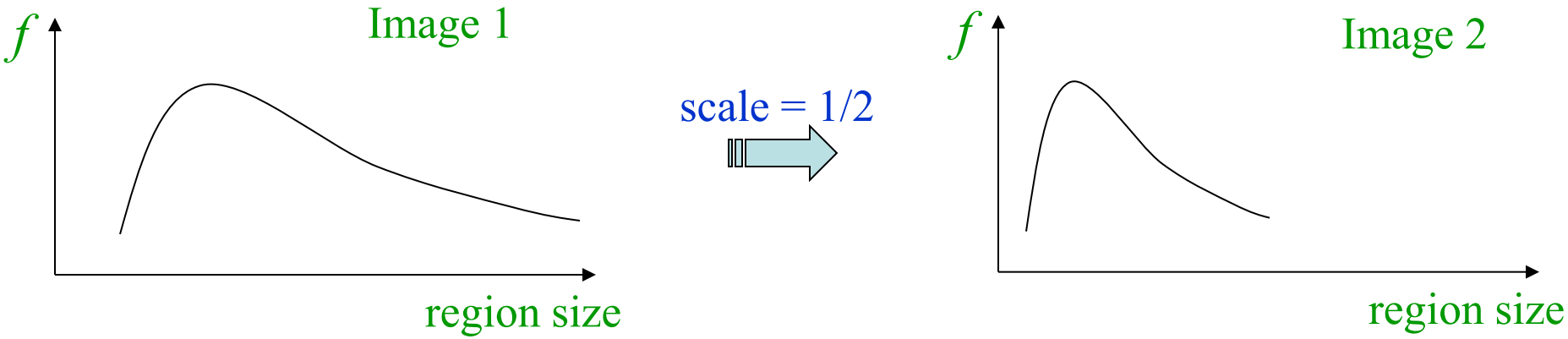


Scale Invariant Detection

- Design this function f on the region (circle), which is "scale invariant" (the same for corresponding regions, even if they are at different scales)

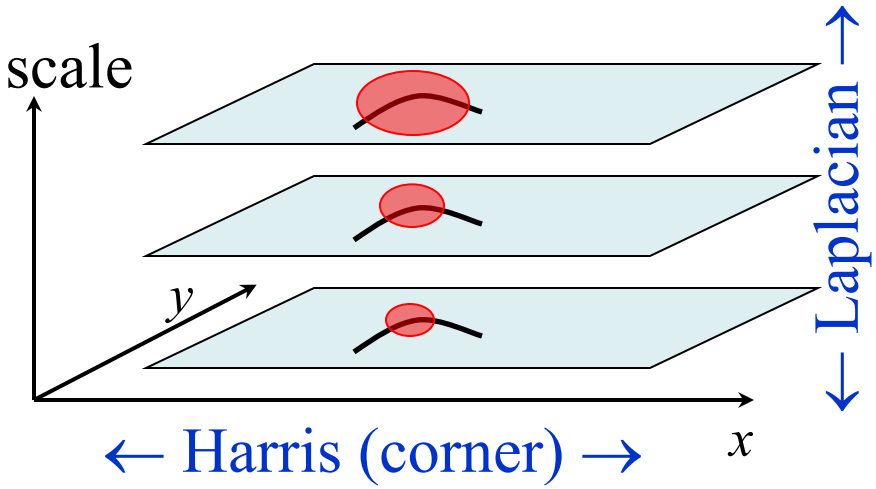
  Example: average intensity. For corresponding regions (even of different sizes) it will be the same.

- At a given location x,y, calculate *f* as a function of region size (circle radius)
- Determine the region size that maximizes *f*
- This determines a landmark-specific scale

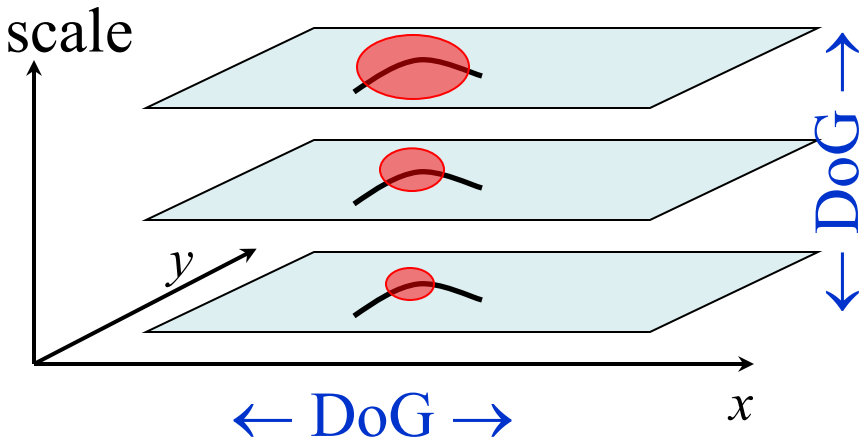


Scale Invariant Detectors

- Harris-Laplacian
  Find local maximum of:
  - Harris corner detector in image space (image coordinates)
  - Laplacian in scale



- SIFT (Lowe)
  Find local maximum of:
  - Difference of Gaussians in space and scale



Topic Overview

Introduction

Feature…

- Desired Characteristics
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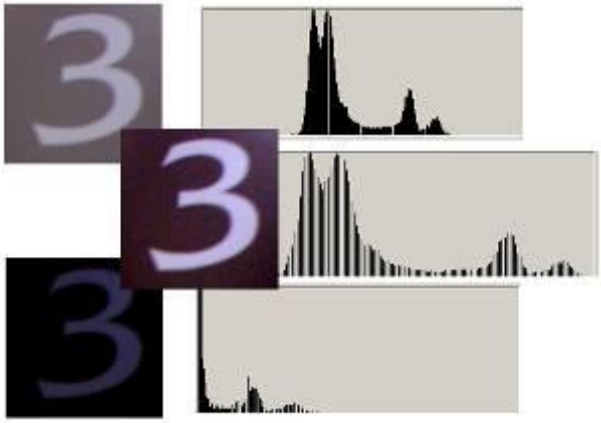
Illumination-Invariant Descriptor

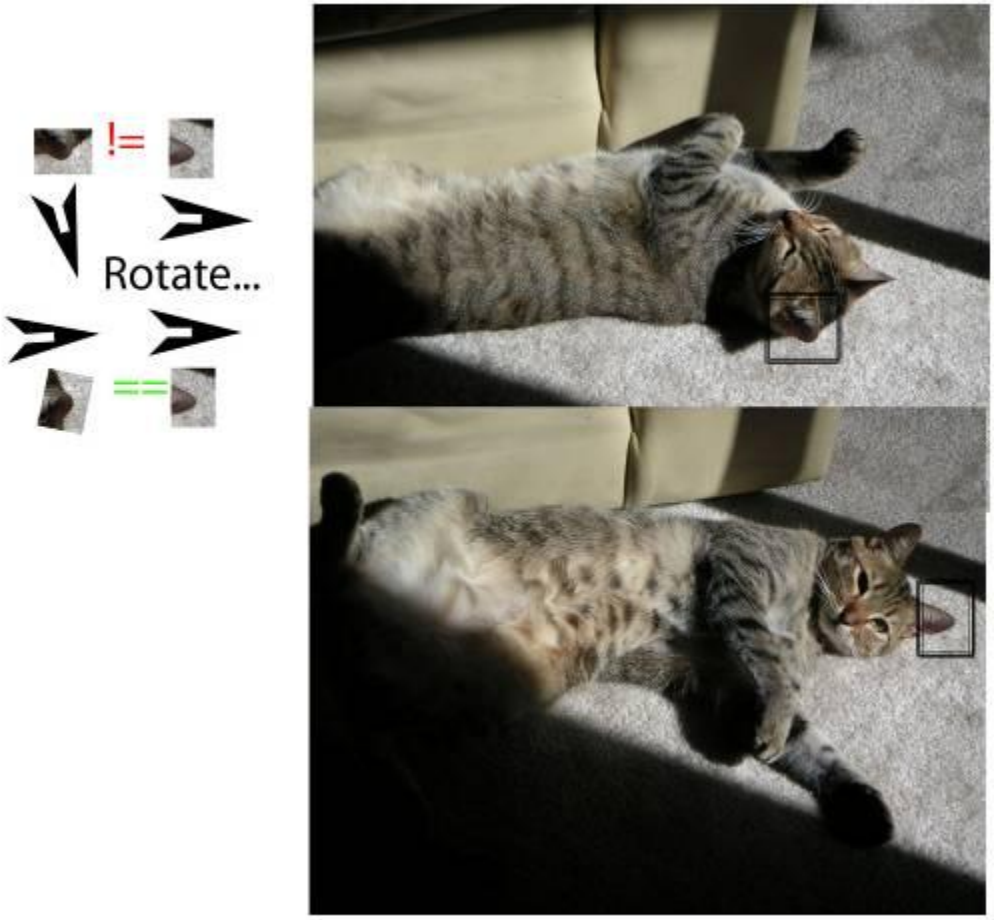
- The easy way (normalized)
- Difference based metrics (random tree, Haar, and SIFT)



Rotation-Invariant Descriptor

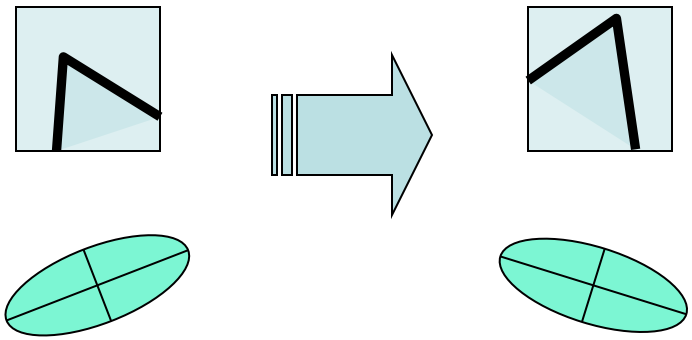
- Rotate all features to go the same way in a deterministic fashion
- Create histogram of gradient directions (SIFT paper: 36; every 10 degrees)
- Rotate to most dominant (maybe second if its good enough, sub-bin accuracy)

Rotation-Invariant Descriptor



Invariance Properties

- Rotation



Eigenvalues do not change -> Invariant

SIFT – Scale Invariant Feature Transform

D.Lowe. “Distinctive Image Features from Scale-Invariant Keypoints”. IJCV 2004

- Descriptor overview:
  - Determine scale (by maximizing DoG in scale and in space), local orientation as the dominant gradient direction.
    Use this scale and orientation to make all further computations invariant to scale and rotation.
  - Compute gradient orientation histograms of several small windows (128 values for each point)
  - Normalize the descriptor to make it invariant to intensity change

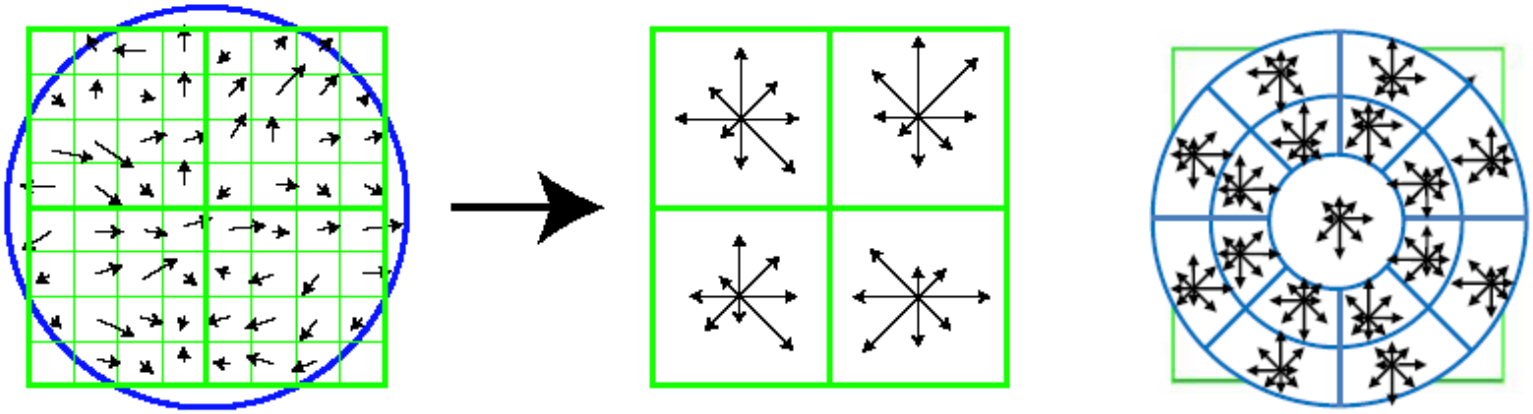


Image gradients

Keypoint descriptor

Topic Overview

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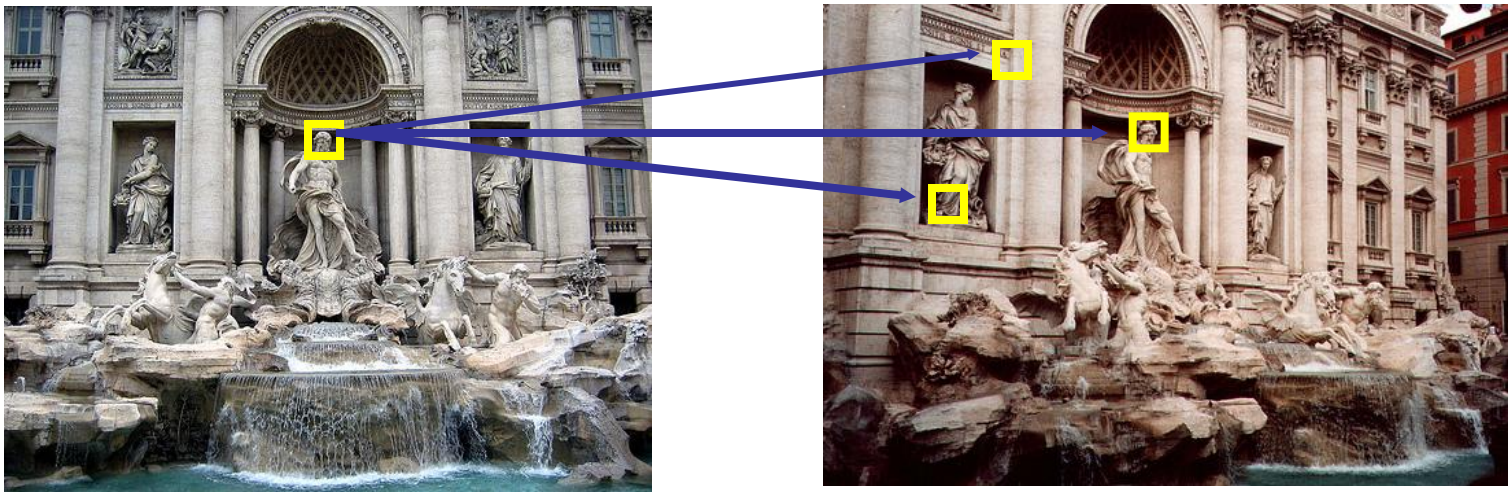
Matching SIFT features

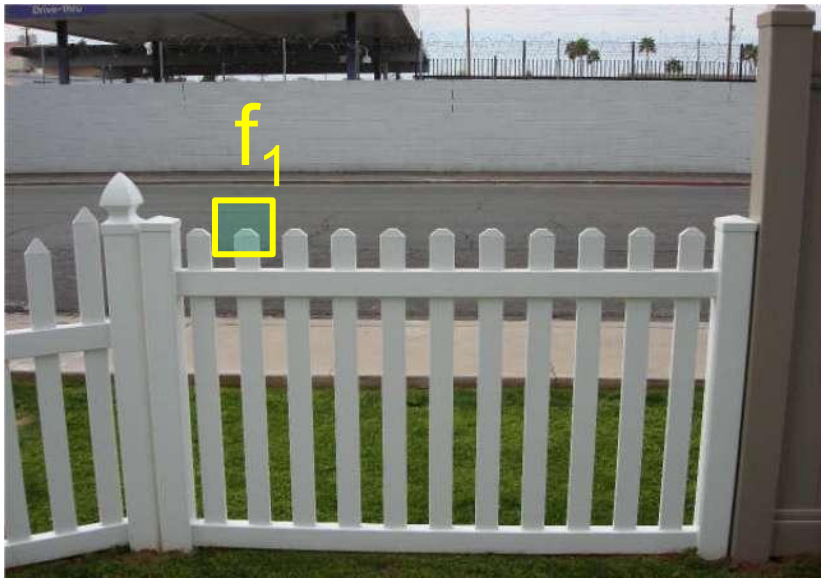
- Given a feature in I_1, how to find the best match in I_2?
  1. Define distance function that compares two descriptors.
  2. Test all the features in I_2, find the one with min distance.



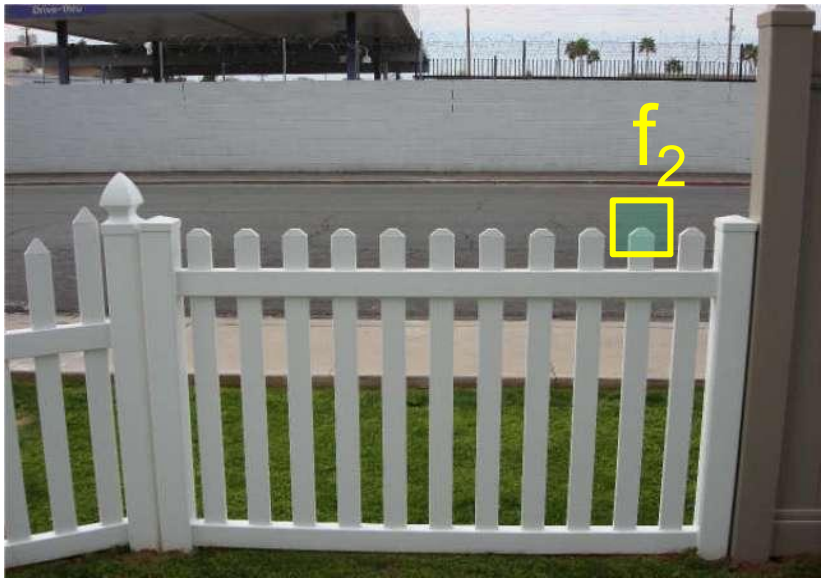
I_1 I_2

From CS485/685 Computer Vision, Dr. George Bebis

Matching SIFT features (cont'd)



I_1

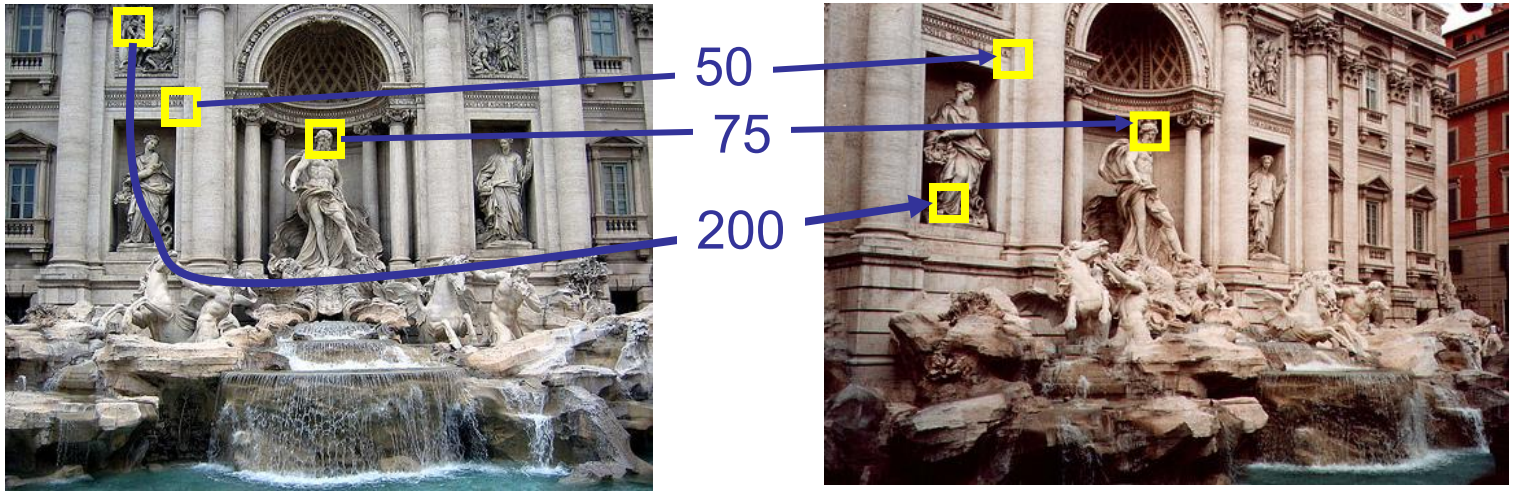


I_2

From CS485/685 Computer Vision, Dr. George Bebis

Matching SIFT features (cont'd)

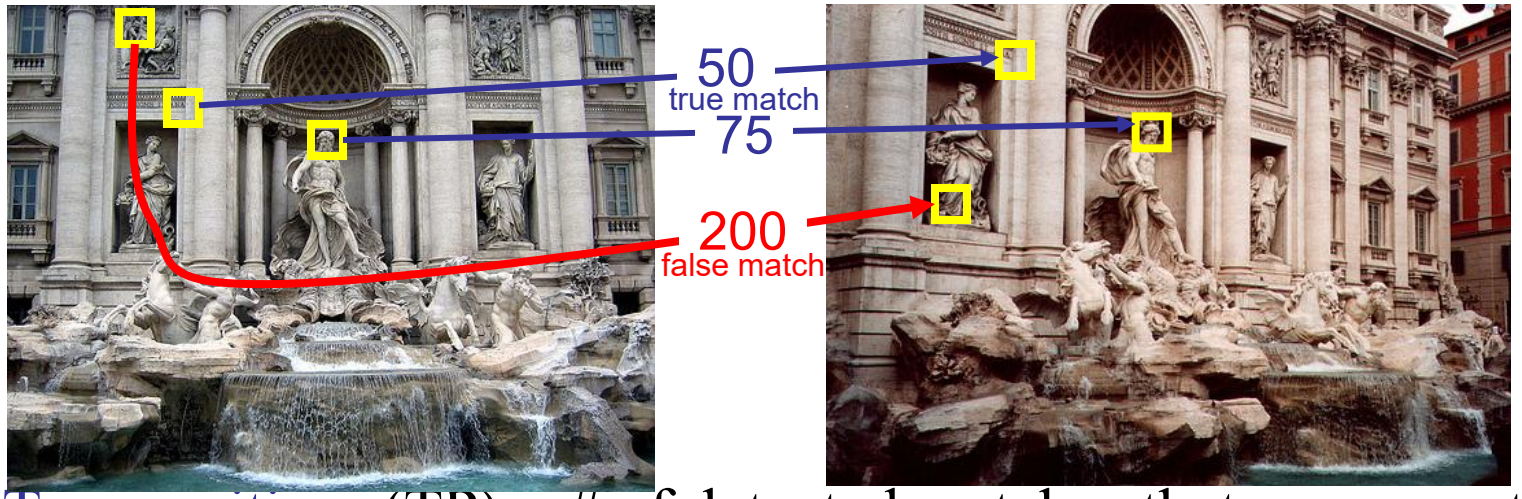
- How to evaluate the performance of a feature matcher?



From CS485/685 Computer Vision, Dr. George Bebis

Matching SIFT features (cont'd)

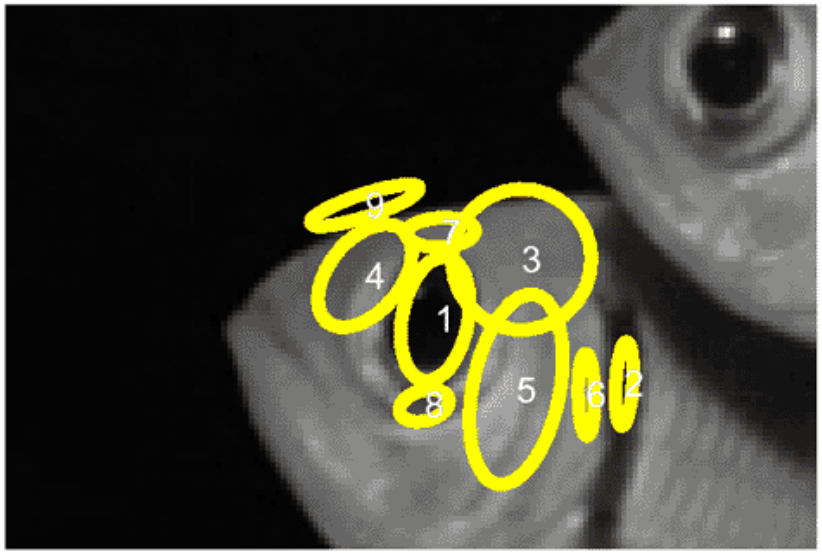
- Threshold t affects # of correct/false matches

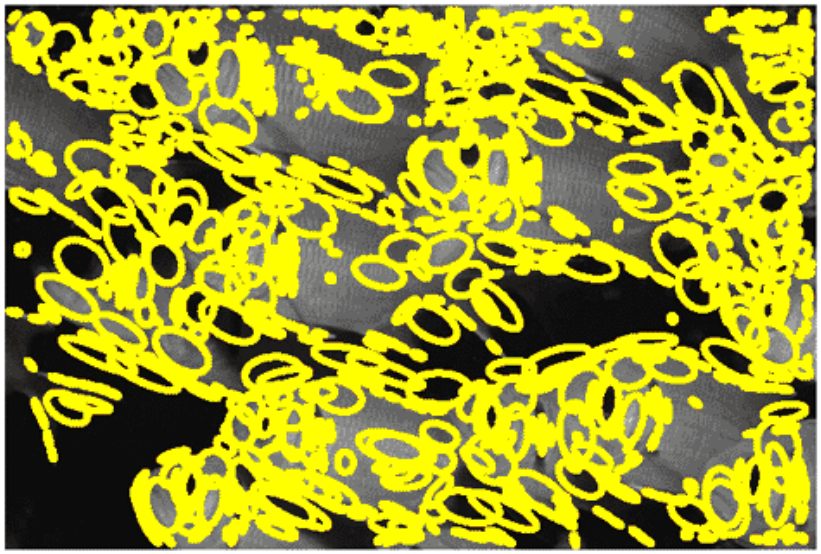


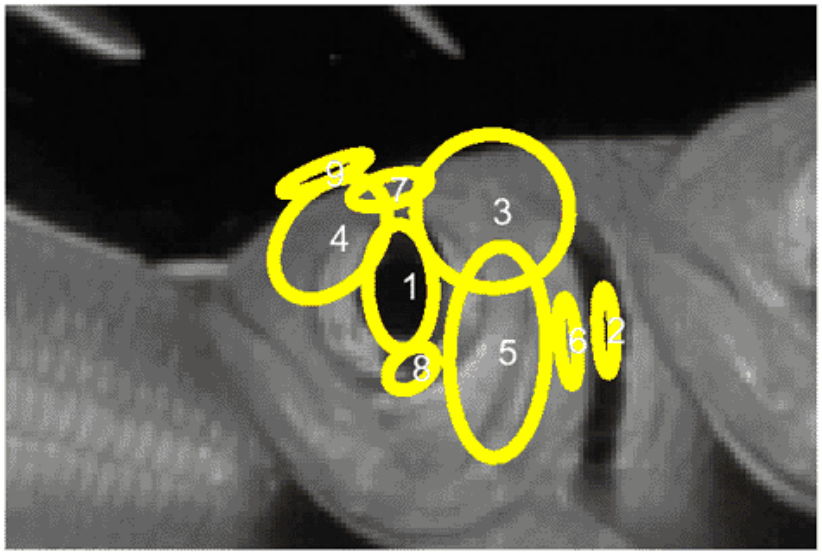
- True positives (TP) = # of detected matches that are correct
- False positives (FP) = # of detected matches that are incorrect

From CS485/685 Computer Vision, Dr. George Bebis

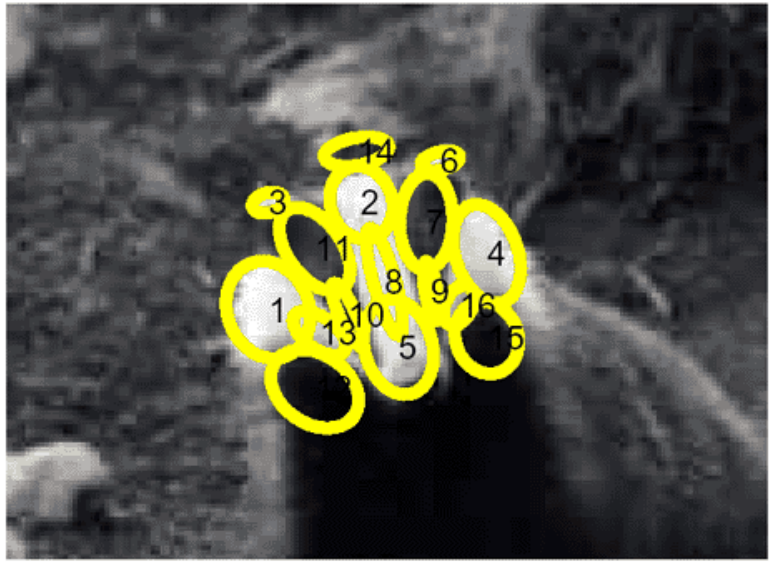
Matching: Finding Repeated Patterns

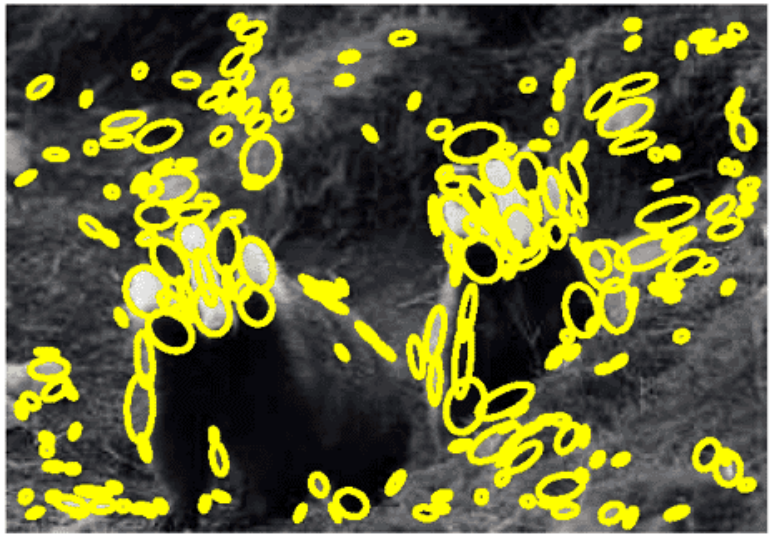


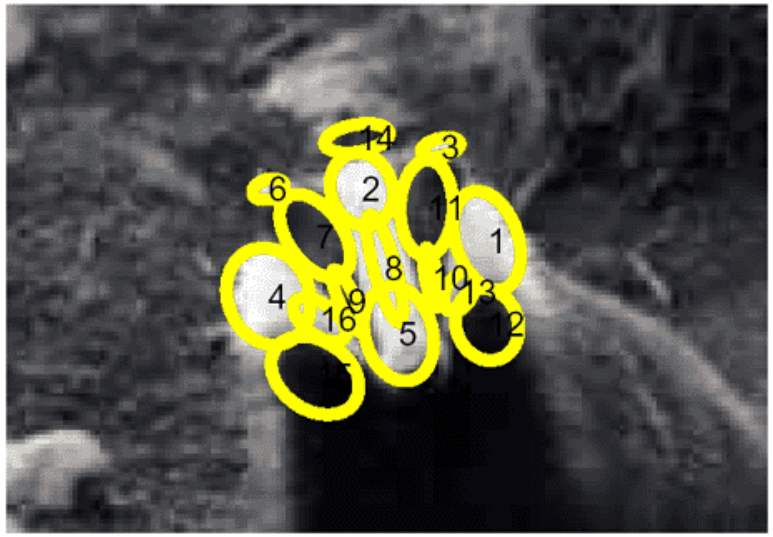




Matching: Finding Symmetries

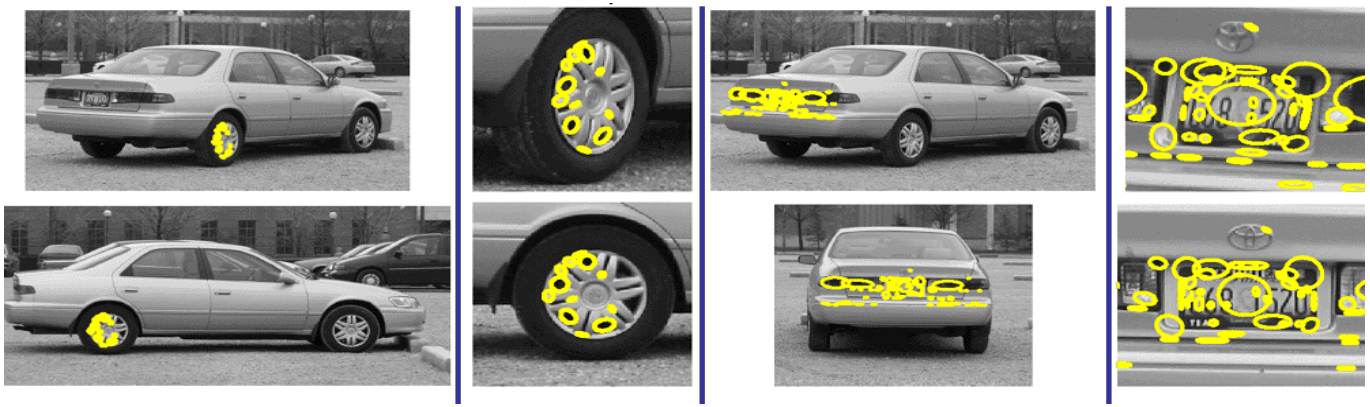






Matching: 3D Objects

closeup



closeup

closeup

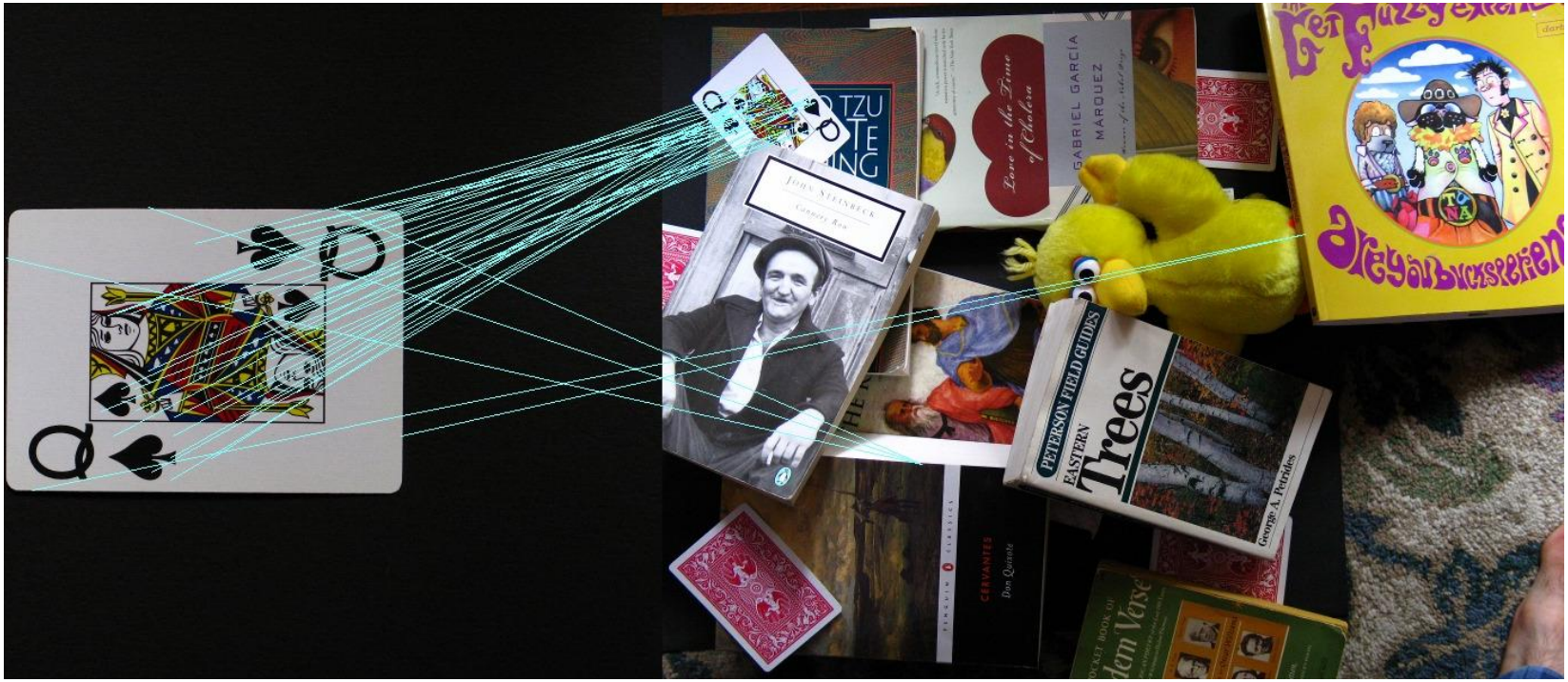
Feature Matching

- Feature space normalization (whitening)
- Matching strategy:

  What's a match?
- Algorithms and implementation:

  How to find a match?
- Image or object-level post-processing
  - Reject outliers
  - Object-match thresholds

Large Object Databases



Features & Tracking in OpenCV

- "Good Features to Track"
  - Shi & Tomasi, 1994
    https://docs.opencv.org/4.10.0/d4/d8c/tutorial_py_shi_tomasi.html
  - Affine distortion model
  - Kanade, Lucas, Tomasi (KLT) tracker: pyramids for fast matching and scale
    https://docs.opencv.org/4.x/d4/dee/tutorial_optical_flow.html
    (note possible confusion with Karhunen–Loève transform)
- Feature detection and description overview
  https://docs.opencv.org/4.x/db/d27/tutorial_py_table_of_contents_feature2d.html
- Feature matching
  https://docs.opencv.org/4.x/dc/dc3/tutorial_py_matcher.html
- Lkdemo
  https://docs.opencv.org/4.10.0/d2/d1d/samples_2cpp_2lkdemo_8cpp-example.html
- Tracking
  https://docs.opencv.org/4.10.0/d5/d0b/group__tracking__detail.html
- Kernelized Correlation Filters (more later)
  https://docs.opencv.org/4.10.0/d2/dff/classcv_1_1TrackerKCF.html